सेंट्रल ट्रांसमिशन यूटिलिटी ऑफ इंडिया लिमिटेड
(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)
(भारत सरकार का उद्यम)
CENTRAL TRANSMISSION UTILITY OF INDIA LTD.
(A wholly owned subsidiary of Power Grid Corporation of India Limited)
(A Government of India Enterprise)

Ref: CTU/N/00/CMETS_NR/21

Date: 28-08-2023

As per distribution list

Subject: 21st Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

Dear Sir/Ma'am,

Please find enclosed the minutes of the 21st Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 31st July, 2023 (Monday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,

28/08/23

(Jasbir Singh)
CGM (CTU)

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Minutes of 21th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 31.07.23

The 21st Consultation Meeting for Evolving Transmission Schemes in Northern Region (CMETS-NR) was held through VC on 31/07/2023. List of participants is enclosed at **Annexure-X**.

A. Confirmation on the minutes of the 20th CMETS-NR meeting

It was informed that the Minutes of the 20th CMETS-NR meeting held on 30/06/2023 were issued vide letter dated 12/07/2023. No comments were received from any applicant, accordingly, minutes were confirmed as circulated. The detailed agenda was discussed as below:

B. Application related matters in Northern Region

Transition of Connectivity/LTA applications in Northern region

It was noted that Hon'ble CERC vide notification dated 01/04/2023 has issued First Amendment of GNA Regulations and notified that the effective date of Regulation 37.1 to 37.8 shall be 05/04/2023 in place of 15/10/2022.

Accordingly, the transition process has been started for all the applicable entities based on their status on 05/04/2023. CTUIL is also receiving fresh applications for Connectivity and GNA under GNA Regulations.

Under the present scenario, CTU is working on Transition process under various provisions of the GNA Regulations, 2022, however, it may be noted that transition is one time activity which requires conversion of all the open access cases of earlier Connectivity Regulations 2009 to new GNA Regulations 2022. As per the Regulations, these cases fall under combination of various categories such as applications for Connectivity (Stage-I&II)/LTA/MTOA with status as effective/granted/applied as on the date of effectiveness of Regulation. For each of the above combinations, different treatments with different timelines have been specified in GNA regulations. Further, during the transition process, large connectivity quantum has been surrendered by the grantees in RE pockets which necessitated re-allocation of the existing/under construction bays/transmission system for optimal utilization. Simultaneously fresh applications are also being received under GNA Regulations.

In dealing with transition cases and new applications under GNA Regulations various issues are faced and the same are being resolved through regular discussions with CERC, CEA, MoP and MNRE. The total process is taking some time and the same has already been communicated to CERC.

The process is now towards the end and the applications received in the month of Jun'23 (**Details enclosed at Table-1**) shall be discussed after completion of transition process as per the priority of applications.

TABLE 1

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought (MW)/date	Nature of Applicant	Generation Schedule	Criterion for applying	Connectivity location (requested)
1.	2200000102	BN Hybrid Power-1 Private Limited	Barmer distt., Rajasthan	01.06.2023	180.8/ 30.06.2025	Generator (Hybrid-Solar:180.8 MW +Wind:70.8 MW)	30.06.2025	Land BG Route	Fatehgarh-IV PS
2.	2200000103	BN Dispatchable-1 Private Limited	Barmer distt., Rajasthan	01.06.2023	300/ 30.06.2025	Generator (Hybrid-Solar:150MW+ Wind:150 MW)	30.06.2025	Land BG Route	Fatehgarh-IV PS
3.	2200000077	Avaada Energy Private Limited	Barmer distt., Rajasthan	05.06.2023	50/ 30.09.2025	Generator (Hybrid-Solar:25 MW +Wind:25 MW)	30.09.2025	Land BG Route	Fatehgarh-IV PS
4.	2200000112	Sprng Energy Private Limited	Bikaner distt., Rajasthan	13.06.2023	200/ 30.06.2025	Generator (Solar)	30.06.2025	Land BG Route	Bhadla-III PS
5.	2200000098	MRS Buildvision Private Limited	Pugal, distt., Rajasthan	14.06.2023	1000/ 30.06.2025	Renewable Power Park developer (Solar)	300 MW-30.06.2025 300 MW-31.07.2025 400 MW-31.08.2025	Land Route	Bikaner-III PS
6.	2200000116	Sprng Energy Private Limited	Barmer distt., Rajasthan	16.06.2023	400/ 31.12.2025	Renewable Power Park developer (Solar)	50 MW-31.12.2025 50 MW-30.06.2026 50 MW-31.12.2026 100 MW-30.06.2027	Land BG Route	Fatehgarh-IV PS

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought (MW)/date	Nature of Applicant	Generation Schedule	Criterion for applying	Connectivity location (requested)
							50 MW-31.12.2027 100 MW-31.12.2028		
7.	2200000111	Sunbreeze Renewables Nine Private Limited	Bikaner distt., Rajasthan	22.06.2023	400/ 01.05.2025	Renewable Power Park developer (Solar)	01.05.2025	Land Route	Bikaner-III PS
8.	2200000110	Renew Solar Power Private Limited	Jodhpur distt., Rajasthan	25.06.2023	400/ 02.01.2025	Generator (Solar)	02.01.2025	LOI	Bhadla-II PS
9.	2200000140	ABREL (RJ) Projects Limited	Barmer distt., Rajasthan	29.06.2023	260/ 31.05.2025	Captive generating plant (Solar)	31.05.2025	Land BG Route	Fatehgarh-III PS
10.	2200000120 (Enhancement)	Helia Energy Park Private Limited	Barmer distt., Rajasthan	30.06.2023	100/ 31.12.2025	Renewable Power Park developer (Solar)	31.12.2025	Land Route	Fatehgarh-IV PS
11.	2200000123	Teq Green Power XV Private Limited	Jaisalmer distt., Rajasthan	30.06.2023	95/ 01.01.2026	Renewable Power Park developer (Solar)	01.01.2026	Land BG Route	Fatehgarh-III PS
12.	2200000118 (Enhancement)	Frugal Energy Private Limited	Jaisalmer distt., Rajasthan	30.06.2023	250/ 15.06.2025	Renewable Power Park developer (Solar)	15.06.2025	Land Route	Bhadla-III PS

C. ISTS Expansion

1. De Linking of Augmentation of 765/400 kV, 1500 MVA transformer at Bhiwani S/s from Transmission system for evacuation of RE power from renewable energy parks in Leh (5GW Leh- Kaithal transmission corridor)

It was deliberated that Transmission system (EHVAC+HVDC) for evacuation of RE power from renewable energy parks in Leh (5GW Leh- Kaithal transmission corridor) was approved in 7th NCT meeting held on 03.12.21. As part of EHVAC system (beyond Kaithal) of above scheme, Augmentation of 765/400 kV, 1500 MVA transformer at Bhiwani S/s (one section has 2x1000 MVA ICT wherein 1500 MVA augmentation will take place, whereas other section has 1x1000 MVA ICT connected through series reactor) along with associated bays incl. 500 MVA spare transformer unit (1- Phase) was also approved. Same was allocated to POWERGRID in RTM vide MOP OM dated 13.01.22 with implementation to be taken up after approval of the Central Government for providing Central Grant for part funding of the project

A meeting of the EFC was held on 3rd June 2022 under the chairmanship of Finance Secretary & Secretary (Expenditure) for appraisal of the proposal of Ministry of New and Renewable Energy (MNRE) on Green Energy Corridor Phase-II - Inter State Transmission system for 13 GW Renewable Energy Projects in Ladakh. In the meeting it was concluded that PIB meeting would be held to take a final view on the proposal after the details have been worked out by MNRE and provided to the DoE including that “The Ministry should re-examine the TBCB mode, especially the feasibility of transmission system in TBCB mode. The Transmission system beyond Kaithal would come up in the plain areas, where not much challenges are envisaged. Accordingly, at the very least the transmission system beyond Kaithal should be considered under TBCB”

In 209th OCC meeting held on 19.07.23, N-1 non-compliance of 765/400kV Bhiwani(PG) ICTs was deliberated. In the meeting it was stated that during high solar generation hours, the loading of 765/400kV 3*1000MVA Bhiwani ICTs is observed to above N-1 loading limit. Maximum ICT loading was observed sometimes 2300MW-2500MW (violated its N-1 loading limit of 2100MW). It was deliberated that Augmentation of 1x1500 MVA (4th) transformer of Bhiwani S/s is linked with Transmission system for evacuation of RE power from renewable energy parks in Leh and with delay in implementation of Leh scheme will also impact the implementation of 765/400kV, 1x1500MVA (4th) ICT at Bhiwani S/s. In view of urgent requirement of 765/400kV, 1x1500MVA ICT, Proposal for delinking of 765/400kV, 1x1500MVA (4th) ICT at Bhiwani S/s from agreed transmission scheme i.e. Transmission system (EHVAC+HVDC) for evacuation of RE power from renewable energy parks in Leh (5GW Leh- Kaithal transmission corridor) will be taken up in Consultation

In Grid India operational feedback report FY 2022-23(Q3/Q4) also, it was mentioned that during high solar generation hours, the loading of 765/400kV, 3x1000MVA Bhiwani ICTs is above N-1 loading limit. Therefore proposal for additional 1500MVA ICT at 765/400kV Bhiwani may be expedited in line with other RE evacuation transmission network in the corridor.

Grid India agreed on proposal and stated that 765/400kV ICT is required on urgent basis. Grid India also stated that in real time, overloading is observed on 765/400kV Bhiwani ICTs which may become n-1 non-compliant. The issue was earlier highlighted in its operational feedback report. CEA also agreed for implementation of augmentation 765/400kV Bhiwani ICT on urgent basis

In the meeting, CTU enquired about Interconnection details for 765/400kV 4th & 5th ICT at Bhiwani S/s. POWERGRID vide mail 31.07.23 informed that Section-I is connected with 02 Nos. 765/400 kV, 1000 MVA ICTs (ICT-2 and ICT-3) and Section-II is connected with 01 No. 765/400 kV, 1000 MVA ICT (ICT-1). Space is available for 765/400 kV, 1500 MVA ICT (4th) (3rd in section-I) & 5th (4th in section-I), however, space is a constraint for Hot spare 500MVA unit for both ICT-4 & ICT-5 but same can be installed as cold spare.

However for ICT-5, at 400kV side, to facilitate the connection in Section-I, GIS Duct is envisaged and dummy dia of Bhiwani(BBMB) shall be utilized. The work involved construction of one no. complete 400kV diameter and shifting of 400kV Bhiwani(BBMB) bay from Bay no. 413 to Dummy side bay.

In view of above, it was agreed that delinking of Augmentation of 765/400 kV, 1500 MVA transformer at Bhiwani S/s along with associated bays incl. 500 MVA spare transformer unit (1- Phase) needs to carried out from Transmission system for evacuation of RE power from renewable energy parks in Leh (5GW Leh- Kaithal transmission corridor) due to early requirement

It was also deliberated that delinking proposal will be taken up in ensuing NCT meeting and based on NCT decision, Augmentation of 765/400 kV, 1500 MVA transformer (4th) at Bhiwani S/s will be taken up as separate scheme with the implementation timeframe of 18 months from allocation. It was also agreed that in case of non-feasibility or delay in delinking of Bhiwani ICT, Augmentation of 765/400 kV, 1500 MVA transformer (5th) (4th in Section-I) at Bhiwani S/s along with associated bays is being taken up for implementation earlier (Implementation time frame 21 months from allocation due to GIB duct) and 4th 765/400 kV, 1500 MVA transformer will be taken up at later stage as part of original scheme i.e. Transmission system for evacuation of RE power from renewable energy parks in Leh (5GW Leh- Kaithal transmission corridor) or as per requirement based on real time loading.

In view of above, following was agreed in the meeting

i. Delinking of Augmentation of 765/400 kV, 1500 MVA transformer along with 500 MVA spare transformer unit (1- Phase) at Bhiwani S/s along with associated bays from the existing scope of Transmission system for evacuation of RE power from renewable energy parks in Leh (5GW Leh- Kaithal transmission corridor)

ii. Separate package will be taken up as ISTS transmission scheme as under
   - Augmentation of 765/400 kV, 1500 MVA transformer at Bhiwani S/s (4th) (3rd in Section-I which have 2x1000 MVA ICTs) along with associated ICT bays
   - 500 MVA spare transformer unit (1- Phase) as a cold spare

Implementation timeframe of 18 months from allocation
Estimated Cost : Rs 165 Crores

2. Non-compliance of N-1 contingency in ICTs at Moga, Malerkotla, Nalagarh and New Wanpoh Substations

It was deliberated that during the 206th OCC meeting, POWERGRID highlighted that presently there is non- compliance of N-1 contingency in ICTs at Moga, Malerkotla, Nalagarh & New Wanpoh Substations of POWERGRID. It was mentioned that at Moga S/S there are two 765/400kV ICT's of 1500 MVA each and Peak loading in ICTs reaches as high as 1900MW (2x950MW) during Oct'2022 to Feb'2023. Further, he intimated that in case 400kV Kishenpur - Moga line is changed to 765kV thereafter these ICT's will become further overloaded.

NRLDC mentioned that they will get this matter examined internally also and requested to take up matter in Consultation Meeting for Evolving Transmission Schemes in Northern Region (CMETS-NR). NRLDC representative stated that Punjab STU views shall also be taken on the cited matter.

ICT augmentation at Nalagarh was already agreed in 17th CMETS-NR meeting held on 31.03.23. Therefore, details of Augmentation of ICTs at Moga, Malerkotla, & New Wanpoh is as under:

A. <u>Augmentation of 765/400 kV, 1x1500 MVA (3rd) ICT at 765/400 kV Moga (PG) substation</u>

It was informed that from the loading pattern of Moga ICTs (2x1500 MVA), it is observed that maximum loading on 765/400kV ICTs is about 1960 MW (980MW/ICT) and loading is higher (>800MW) in winter season (Oct'22-Feb'23). From studies, it is emerged that on 2 nos. of 1500 MVA ICTs, loading of ICTs was more than 'N-1' compliance limit (>870 MW) in solar maximized scenario of winter season at few instances. Further from rolling plan studies (2027-2028) also, ICT loadings are of similar order. The loading pattern of Moga ICTs for past one year is as below (Fig 1):

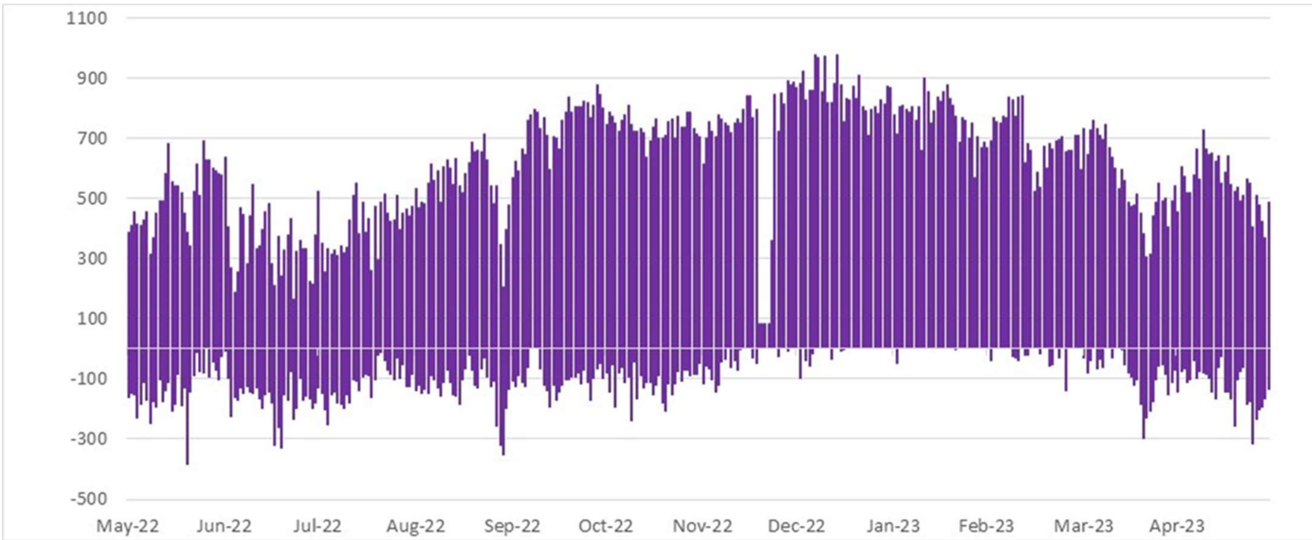

Fig 1: Loading of Moga ICTs of Past one year (Source: Grid-India)

POWERGRID vide mail dated 28.06.23 confirmed space availability for Augmentation of 765/400kV transformer (3rd 1500MVA) along with transformer bays at 765/400kV Moga S/s. 765KV and 400kV side bay equipment along with Bus connection for both voltage level shall be GIS type. It was also informed that 4x500MVA single phase ICT may be considered, as existing spare ICT is difficult to utilitse as spare with proposed ICT.

During the meeting, Grid India indicated that at present only for some instances, the ICT loading at Moga S/s is more than 800 MW and presently, ICT augmentation is not critically required at Moga, however, in case, in the Rolling Plan reports of CTUIL for 2027-28 scenario, if such need emerges, then this augmentation may be taken up for implementation.

Further, PSTCL enquired about the effect of Transmission charges on STU with this augmentation scheme. On this, CTUIL, informed that transmission charges shall be applicable as per the Sharing Regulations depending upon the category defined for this ICT i.e. RE Component/Non-RE Component. CTUIL also clarified that overloading at Moga S/s occurs mostly at peak solar hours and therefore, this augmentation may be categorized under RE Component as it is being utilized in Peak Solar hours. CTU stated that from last one year data it emerged that overloading happens only for 1.3% times (>800MW) and 0.4% times (>850MW) and in CTU studies also loadings are of similar order in 2027-28 scenario and accordingly, views from all constituents may be taken up.

CEA opined that loading of Moga ICT is higher for very limited time duration and therefore augmentation may be kept on hold and to be monitored in real time. CEA also opined that no. of hydro projects are coming in Kishenpur/Kishtwar complex which may necessities charging of Kishenpur-Moga 400 kV D/c line on 765 kV level and accordingly ICT requirement may be reviewed. CTU stated that studies are undergoing for hydro projects in J&K/HP and Prima facia upgradation requirement of Kishenpur-Moga 400 kV D/c line along with Moga ICT augmentation is not envisaged in studies due to high solar injection on Moga S/s.

CTU requested Grid-India to monitor ICT loading in future and suggested that based on operational feedback and real time loadings indicated by Grid-India, ICT augmentation requirement will be reviewed. Grid-India agreed for the same and stated that in future if SPS may also be considered if ICT loading may increases suddenly.

CTUIL deliberated that in future, POWERGRID may discuss technical agendas in consultative meeting (CMETS) meeting first before posing it to OCC/NRPC forum and based on technical deliberation in CMETS meeting, same may be forwarded to requisite forum. CTU requested NRPC not to consider such agendas directly without any deliberation in consultative meeting. NRPC agreed for the same. Accordingly, it was decided in the meeting that Augmentation of 765/400 kV, 1500 MVA (3rd) ICT at Moga S/s may be kept on hold at present and based on real time loadings indicated by Grid-India and operational feedback reports, same may be reviewed.

B. <u>Augmentation of 400/220 kV, 1x500 MVA (3rd) ICT at 400/220kV New Wanpoh substation</u>

It was deliberated that at present, 400/220kV New Wanpoh Substation has 2x315MVA ICTs (Single phase units of 105MVA) ICTs and New Wanpoh is interconnected with Mirbazar S/s through 220kV D/c line. In addition to that, 2 nos. 220 kV interconnections i.e. 220 kV New Wanpoh – Alusteng D/c Line & 220 kV New Wanpoh – Mattan D/c Line is being implemented by JKPTCL for which 220kV bays are already commissioned.

In the 204th OCC meeting held on 17.02.23, Status was updated by JKPTCL for above 220kV lines, which is as under

- 220 kV New Wanpoh – Alusteng D/c Line: The work is in progress and expected to be commissioned by the end of 2023.

- 220 kV New Wanpoh – Mattan D/c Line: the funding source for the project is being identified and the project is expected to be completed by ending 2024.

From the loading pattern of New Wanpoh ICTs (2x315 MVA), it is observed that Loading of ICTs have reached maximum of about 390 MW (195MW/ICT) in past one year and loading is not breaching N-1 limit for peak Loading of ICTs (460 MW i.e. 230MW/ICT). The loading pattern of New Wanpoh ICTs for past one year is as below (Fig 2)

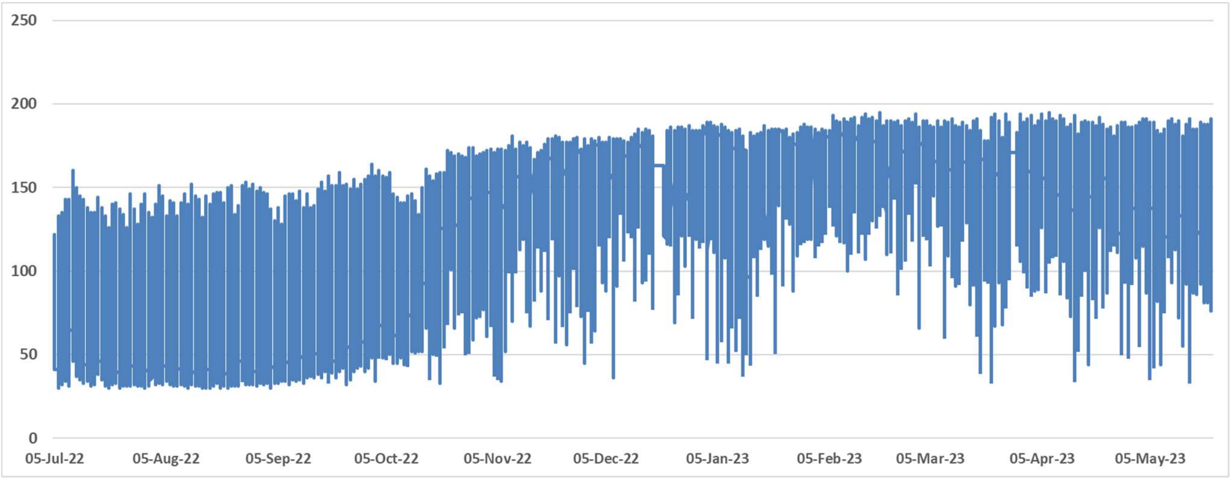

Fig 2: Loading of New Wanpoh ICTs of Past one year (Source: Grid-India)

Further, from rolling plan studies (2027-28) also, ICT is not breaching N-1 limits (after considering future interconnections). JKPTCL was requested to provide their inputs on 400/220kV ICT requirement at New Wanpoh S/s w.r.t upcoming interconnections from Mattan and Alusteng S/s along with details of load growth in above area. JKPTCL was also requested to update the commissioning schedule of above 220kV lines. POWERGRID vide mail dated 28.06.23 confirmed the space availability for Augmentation of 400/220kV transformer (3rd 315 MVA Single phase units of 105MVA).

JKPCTL stated that at present Alusteng substation is interconnected to Zainkote and Leh through 220kV lines and to provide redundant power supply, interconnection from New Wanpoh to Alusteng is under implementation and may delay for 1-2 months.

JKPTCL stated that at present there is no need of this additional ICT, however, with the commissioning of their Mattan & Harven substations (to be commissioned in sep'23 with interconnection with Alusteng s/s), the loadings will increase at New Wanpoh S/s and therefore ICT augmentation may be considered in pipeline for implementation. CTU enquired about schedule of 220 kV New Wanpoh – Mattan D/c Line. JKPTCL informed that New Wanpoh -Mattan line has funding issues and implementation may take minimum 18 month's time from present. CTU stated that as per sharing regulation, charges will be borne by JKPTCL in transformer component and therefore they need to confirm the ICT requirement at Ne Wanpoh along with schedule as ICT augmentation may take minimum 18 months.

POWERGRID (NR-2) suggested that implementation time of 18 months may not be sufficient for new ICT installation at New Wanpoh S/s due to transportation and limited working window in winter season. CTU noted the same and sated that special consideration for ICT implementation timeline will be decided at allocation stage.

Grid-India suggested that ICT augmentation can be taken up as ICTs may become n-1 non compliant in future. and it is suggested that JKPTCL may agrees to take up the ICT augmentation. JKPTCL informed that they are expediting interconnection of New Wanpoh – Alusteng line and in view of growing demand and provide redundancy to Alusteng, JKPTCL require ICT augmentation at New Wanpoh substation. CTU stated that JKPTCL may also confirm the requirement to CTU through mail.

Subsequently, JKPTCL vide letter dated 21.08.23 to CEA informed that the transmission plan (Kashmir region) for 2022-27 envisages addition of 2479 MVA at different voltage levels and stands approved in the CEA meeting held on 19-02-2021. This includes 1130 MVA at 220/33 kV Level, 480 MVA at 220/132 kV level and 869 MVA at 132/33 kV level. The commissioning of the new Grid Stations necessitates provision of new sources or capacity augmentation at 400/220 kV level to enable the trickling down of the benefits of the new Grid stations to the consumers. In the JKPTCL letter, it was also stated that the existing sources of power shall not be able to cater to the increased demand of power and this necessitates immediate steps to augment the sources. The 220/132kV GSS Delina has been augmented from 320 MVA to 480 MVA and it is proposed to augment the existing and 220/132 kV GSS Zainakote from 450 MVA to 700 MVA by the 2024.

In the letter it was also informed that maximum drawl of power from 400/220kV, 2x315MVA (630MVA) during the previous winters has been more than 400MW and as such it is essential that New Wanpoh substation is augmented by atleast 50% i.e. 315MVA to meet the growing demand of power in the valley **(copy of letter is attached in Annexure-I)**

In view of above following was agreed in ISTS is as under

- Augmentation of 400/220 kV,1x315 MVA (3^{rd}) ICT (3x105 MVA single phase units) at New Wanpoh S/s along with associated transformer bays

C. <u>Augmentation of 400/220 kV, 1x500 MVA (4th) ICT at 400/220 kV Malerkotla substation</u>

It was deliberated that from the loading pattern of Malerkotla ICTs (2x315+1x500 MVA), it is observed that Loading of ICTs have reached maximum of about 830 MW (on 1130 MVA) in past one year (Jun'22-Aug'22) in Paddy season of Punjab and reaching to N-1 limit level for peak Loading of ICTs (~830 MW). The loading pattern of Malerkotla ICTs for past one year is as below (Fig 3):

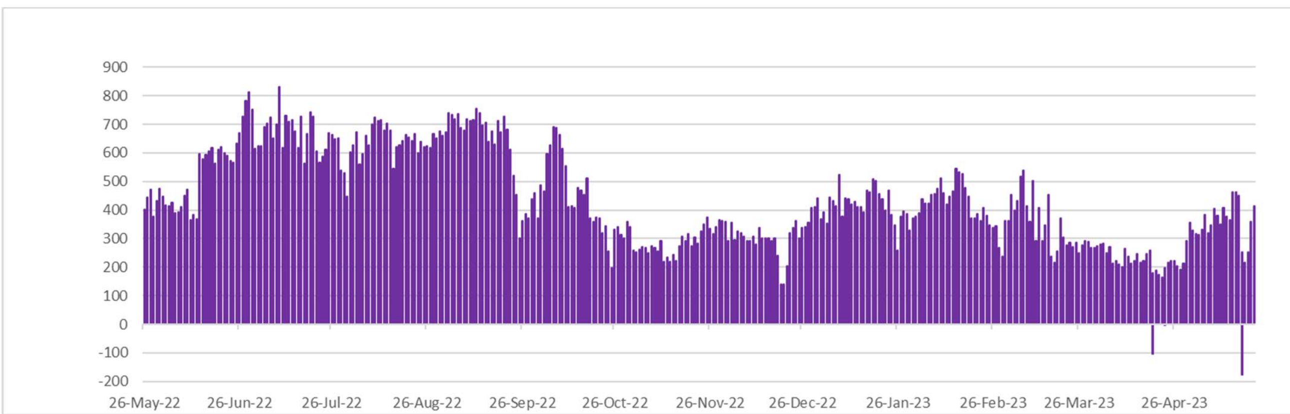

Fig 3: Loading of Malerkotla ICTs of Past one year (Source: Grid-India)

Further from rolling plan studies (2027-28) also, ICT loading (~880MW) breaching N-1 limits increasing in Paddy season load of Punjab.

POWERGRID vide mail dated 28.06.23 confirmed the space availability of 400/220kV transformer (4^{th} 1x500MVA) along with transformer bays at 400/220kV Malerkotla S/s (existing ICTs 2x315+1X500MVA). It was informed that 400kV side bay equipment shall be GIS type and 220kV side shall be routed to PSTCL yard through 220kV GIB/Cable. 220kV side ICT bay space may be confirmed from PSTCL end. Further CTU vide mail 17.07.24 requested PSTCL to provide their observations on requirement of ICT at Malerkotla S/s w.r.t. present and future demand. It was also requested that PSTCL shall provide details regarding space availability for 220kV side ICT bay at Malerkotla Substation.

In the meeting, PSTCL stated that space is available and same was communicate to POWERGRID with 3-4 options along with most suitable option. CTU requested to provide the above SLD, which was shared by PSTCL on 31.07.23. PSTCL enquired about techno-economics of GIB/Cable. POWERGRID informed that based upon RoW issue and hindrance, the selection could be made, however, cable would be cheaper than GIB if both the options are feasible.

PSTCL confirmed the requirement of ICT at Malerkotla S/s w.r.t. present and future demand. GRID India stated that ICT augmentation at Malerkotla may be considered based on PSTCL requirement, however, recently augmentation has been done at Patiala and this may affect ICT loadings at Malerkotla too.

PSTCL again confirmed that ICT shall be required in paddy season and ICT loadings have been breached at Malerkotla for some instances in present Peddy season also despite of Patiala ICT augmentation. PSTCL also stated that as ICT implementation may also take about 2 years, same may be approved now so that ICT would be available by 2025.

CTU stated that for ICT augmentation may require 21 months for implementation as it involves 400kV side bay equipment of GIS type. Regarding 220kV side bay along with GIB/cable, PSTCL requested that 220kV works may be implemented in ISTS by same implementing agency assigned for transformer augmentation for better coordination, however, if 220kV work is assigned to PSTCL, 21 months time is sufficient for 220kV bay implementation. CEA also agreed for augmentation requirement at Malerkotla S/s. In view of that it was decided that ICT augmentation along with 400kV & 220kV side bay equipment is to be implemented in ISTS

Scope of work agreed in ISTS is as under

- Augmentation of 400/220 kV, 1x500 MVA (4^{th}) ICT at Malerkotla S/s along with associated transformer bays
- 400kV side bay equipment shall be GIS type
- 220kV side shall be routed to PSTCL yard through 220kV GIB/Cable

3. **<u>Requirement of one set of Bus Sectionalizer at 400kV level of 400/220kV Bikaner-II PS</u>**

It was informed that Transmission system strengthening scheme for evacuation of power from solar energy zones in Rajasthan (8.1 GW) under Phase-II" was discussed in 4th meeting of NRSCT held on 25.07.2019 and recommended for implementation through TBCB in the 6th meeting of erstwhile NCT held on 30.09.2019. As part of above , Ph-II, Part F scheme was modified in 4th NCT meeting held on 20.01.20 and 28.01.20 (deletion of 400/220 kV ICTs at Bikaner-II PS from approved scope).

Subsequently 2 nos. of 500MVA ICTs (1st & 2nd) and 1 no. of 500 MVA ICT (3rd) at Bikaner-II PS was allocated in RTM to M/s POWERGRID Bikaner Transmission System Limited (PBTSL) for implementation vide CTU OM dated 16.11.21 and 08.06.23 respectively. Further, Augmentation by 400/220 kV, 5x500 MVA ICT (4th to 8th) at Bikaner-II PS was agreed for implementation as part of transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-1) (Bikaner Complex)-Part E and same was allocated to M/s PBTSL vide CTU OM dated 15.11.22 (based on NCT letter dated 15.11.22). As per above NCT letter, above ICTs were to be taken up for evacuation requirement beyond 2000 MW at 220 kV level of Bikaner-II PS, with implementation timeframe matching with schedule of RE generation or 18 months from date of allocation, whichever is later. Subsequently, CTU vide mail 13.06.23 informed to POWERGRID for take up implementation of 5x500 MVA ICTs (4th to 8th) at Bikaner-II PS with implementation schedule of Dec'24 (4th to 7th) and Jan'25 (8th) after meeting the requirement specified in NCT letter dated 15.11.22.

POWERGRID vide letter dated 27.06.23 informed that 400kV bus sectionalizer (bays 428 &429) at Bikaner-II PS may be required after ICT-5 for taking up implementation of ICT-6th to 8th. It was also informed that in absence of 400kV bus sectionalizer bays, integration of existing 400kV section bus bar protection system & new bus bar protection system shall not be feasible. In view of the above, one set of sectionalizer at 400kV level of 400/220kV Bikaner-II PS is required.

Further, it was deliberated that tie one no. of 400kV tie bay (associated with 400/220kV ICT-4) shall also be required.

In view of above deliberations, following ISTS scheme was agreed:

- One set of bus sectionalizer at 400kV level of 400/220kV Bikaner-II PS
- One no. of 400kV tie bay (associated with 400/220kV ICT-4 as per approved SLD)

Implementation Timeframe: Matching with implementation of 400/220kV ICTs (4th to 7th) i.e. Dec'24

4. **<u>Augmentation of 5th ICT at Fatehgarh-III PS(Section-1)</u>**

It was deliberated that Transmission System Strengthening for potential solar energy zones Phase-II in Northern Region for 8.1 GW was discussed and agreed in the 5th Meeting of Northern Region Standing Committee on Transmission (NRSCT) held on 13.09.2019. Subsequently, the scheme was approved in the 6th meeting of National Committee on Transmission (NCT) held on 30.09.2019. As part the above scheme, Ph-II Part-A involved establishment of 4x500 MVA Ramgarh PS (now Fatehgarh-III PS (sec-1)) along with 400 kV D/c lines towards Fatehgarh-II PS & Jaisalmer-II (RVPN) S/s.

Subsequently, as part of Transmission system for evacuation of power from REZ in Rajasthan under Phase-III(20GW), augmentation of 5th ICT at Fatehgarh-III PS(Sec-1) was discussed and agreed in the 3rd NRPC(TP) held on 19.02.2021. Further, the 5th ICT at Fatehgarh-III PS(Sec-1) was taken up for discussion and approval in the 5th NCT meeting held on 25.08.2021 and 02.09.2021 as part of Ph-III Part-J scheme.

However, in the 5th NCT meeting, it was opined that one of the pilot project of Battery Storage System of 250 MW is being planned at perspective location near Fatehgarh-3, therefore there might be a possibility that the 5th ICT of 500 MVA at Fatehgarh-III PS(Sec-1) may not be required and hence this item needs to be reviewed. Accordingly, the Augmentation of ICT(5th) at Fatehgarh-III PS (Sec-1) was deferred in the 5th NCT meeting.

Subsequently, in the manual on Transmission Planning Criteria 2023 by CEA published in March 2023, it is mentioned that 'N-1' reliability criteria may be considered for ICTs at the ISTS /STU pooling stations for renewable energy based generation of more than 1000 MW after considering the capacity factor of renewable generating stations.

Cumulative connectivity quantum of 2280 MW was granted at Fatehgarh-III PS(Sec-1). Recently, 300 MW has been surrendered in GNA Transition process. The remaining connectivity granted at Fatehgarh-III PS is 1980 MW. Therefore, to satisfy N-1 reliability criteria at 400/220 kV Fatehgarh-III PS(Section-1), it is necessary to take up augmentation of 5th ICT at Fatehgarh-III PS(Sec-1) is required.

Further, in the meeting for reallocation of connectivity bays in NR held on 20.06.2023, it was decided that in order to ensure N-1 compliance as per planning criteria 2023, the 220 kV line bay which is surrendered under GNA transition(300 MW) at Fatehgarh-III PS(Sec-1) shall not be granted to any other RE applicant and instead shall be utilized by upcoming pilot BESS project (2x250 MW) at Fatehgarh-III PS which is under implementation M/s JSW. However, despite interconnection of BESS, for N-1 compliance there is a need of 400/220 kV 500 MVA 5th ICT at Fatehgarh-III PS(Sec-1).

The Fatehgarh-III PS(Section-1) is currently under advanced stage of implementation by POWERGRID(TBCB) & is expected to be commissioned progressively from Aug'23. As per the latest JCC minutes,1980 MW of generation projects granted connectivity at Fatehgarh-III PS(Section-1) are expected to be commissioned progressively from Sep'23. In view of the above, implementation of 5th ICT at Fatehgarh-III PS(Sec-1) is to be taken up on urgent basis to meet the N-1 compliance as per CEA Transmission Planning Criteria 2023.

In meeting SECI queried that requirement of Augmentation of 5th ICT at Fatehgarh-III PS(Section-1) is due to BESS. CTU stated that this ICT augmentation is not for BESS , it is for requirement to meet the N-1 compliance as per CEA Transmission Planning Criteria 2023. SECI also queried regarding 220kV line bays for this ICT augmentation, CTU stated that for this ICT augmentation requires separate transformer bays. In view of above deliberations, following ISTS scheme was agreed.

SECI enquired about revision of RE capacity of 1980MW from earlier granted 2280MW capacity at Fatehgarh-III PS(Section-1). CTU stated that earlier 2280MW connectivity allocated to RE applicants at Fatehgarh-III PS(Section-1) against potential of 1.98GW in Ph-II. However, in view of surrendered capacity (300MW) at Fatehgarh-III PS(Section-1) and new N-1 compliance requirement as per CEA Transmission Planning Criteria 2023, RE connectivity at Fatehgarh-III PS(Section-1) allocated to1980MW.

In view of above deliberations, following ISTS scheme was agreed:

- Augmentation of 400/220 kV ,1x500 MVA (5th) ICT at Fatehgarh-III PS(Section-1) along with transformer bays

Annexure-X

List of Participants of 21st Consultation meeting for Evolving Transmission Schemes in NR held on 31.07.2023

CEA

Shri Nitin Deswal	Deputy Director
Shri Kanhaiya Singh	Assistant Director

SECI

Shri Kaustuv Roy	GM
Shri R.K.Agarwal	Consultant

Grid India

Shri Alok Kumar	GM
Shri Gaurav Malviya	Manager
Shri Gaurab Dash	Assistant Manager

NRPC

Shri Pradeep Kumar	Executive Engineer
Shri Rajat Dixit	Assistant Executive Engineer
Shri Reeturaj Pandey	EE & Assistant Secretary

CTU

Shri Kashish Bhambhani	GM (CTU)
Shri Sandeep Kumawat	DGM (CTU)
Smt. Ankita Singh	Ch. Manager (CTU)
Shri R Narendra Sathvik	Manager (CTU)

Shri Madhusudan Meena	Engineer (CTU)
POWERGRID	
Shri Abhay Kumar	CGM
Shri Ajit Kumar	CGM
Shri Jagat Ram	GM
Shri Gautam Luthra	Sr DGM (CMG)
Shri Rakesh Gupta	Ch. Manager
HVPNL	
Shri Sanjay Kumar Arora	Chief Engineer (PD&C)
Shri Sanjay Verma	SE (Planning)
Shri Deepak Sarit	XEN (System Study)
Smt Palak Sinha	AE (System Study)
PSTCL	
Shri Vivek Khanna	Dy. CE
Shri Nitin Kumar	Sr. XEN/Planning-1
Shri Surjeet Sood	
PTCUL	
Shri Lalit Kumar	SE
Shri Ashok Kumar	Executive Engineer
JKPTCL	
Mr. Ehtisham Andrabi	

Jammu & Kashmir Power Transmission Corporation Ltd.
Office of the Chief Engineer, JKPTCL, Kashmir,
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The Chief Engineer,
Power Survey and Load Forecasting,
CEA, New Delhi.

No: CE/JKPTCL Kmr/6100-6103 Dated: 21-08-2023

Sub: Augmentation of 630 MVA, 400/220 kV GSS New-Wanpoh.
Sir,

It has been observed that the demand of Power in Kashmir Valley is increasing rapidly over the past few years and the maximum restricted demand of about 1900 MW was recorded last winter. As already intimated in various meetings, the transmission plan (Kashmir region) for 2022-27 envisages addition of 2479 MVA at different voltage levels and stands approved in the CEA meeting held on 19-02-2021. This includes 1130 MVA at 220/33 kV Level, 480 MVA at 220/132 kV level and 869 at 132/33 kV level. The commissioning of the new Grid Stations necessitates provision of new sources or capacity augmentation at 400/220 kV level to enable the trickling down of the benefits of the new Grid stations to the consumers. The existing sources of power shall not be able to cater to the increased demand of power and this necessitates immediate steps to augment the sources. The 220/132 kV GSS Delina has been augmented from 320 to 480 MVA and it is proposed to augment the existing 220/132 kV GSS Zainakote from 450 MVA to 700 MVA by the 2024.

The maximum drawal of power from 630 MVA GSS New-Wanpoh during the previous winter has been more than 400 MW and as such it is essential that PGCIL owned GSS New-Wanpoh is augmented by atleast 50% i.e., 315 MVA to meet the growing demand of Power in the valley. It is in place to mention that 400 kV incomer Transmission Line for the Grid Sub-Station is having twin Moose conductors and can carry the extra load safely.

It is once again requested to take up the matter with the concerned quarters to enable planning of the Transmission facilities accordingly.

Yours Sincerely,

Chief Engineer

Copy to the:
1. Principal Secretary JKPDD Jammu for information.
2. Managing Director JKPTCL Jammu for information.
3. The COO, CTUIL, PGCIL New-Delhi for information.